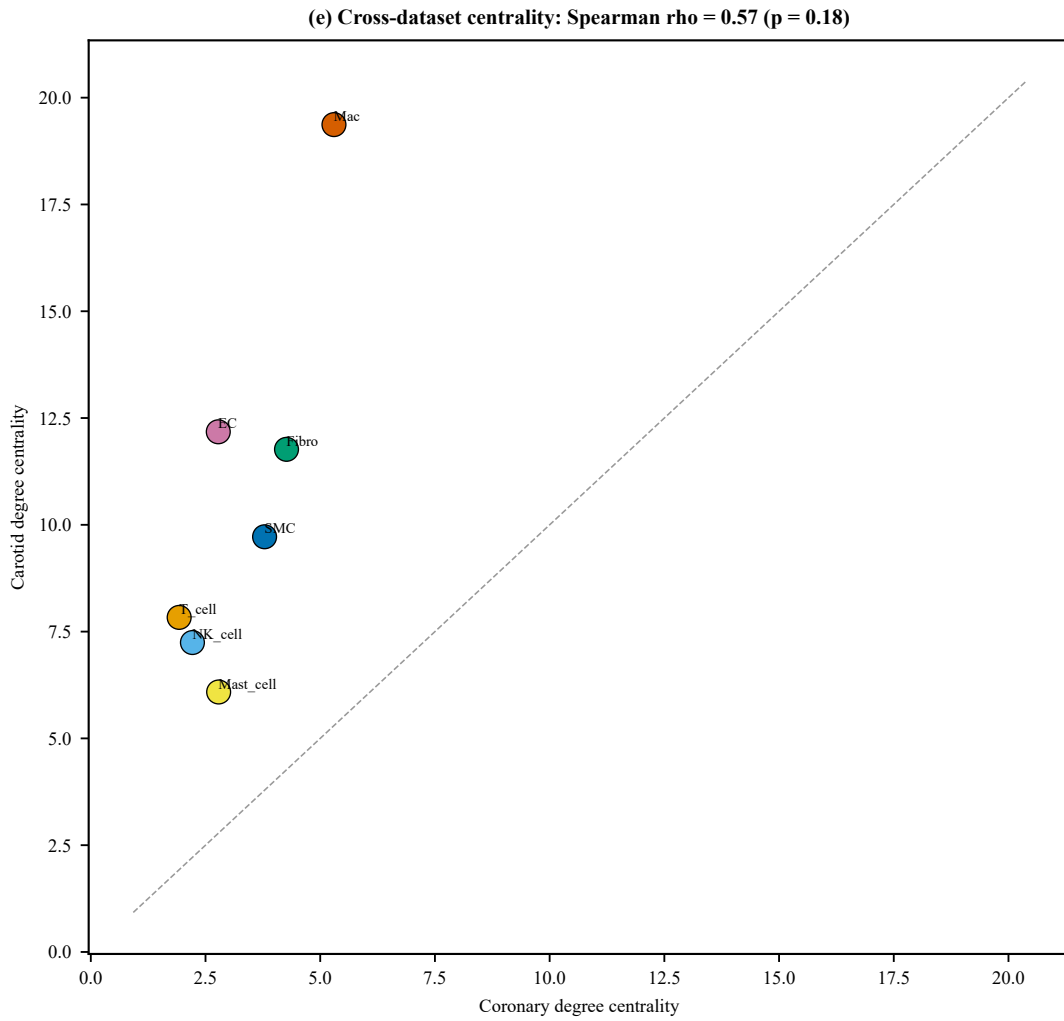
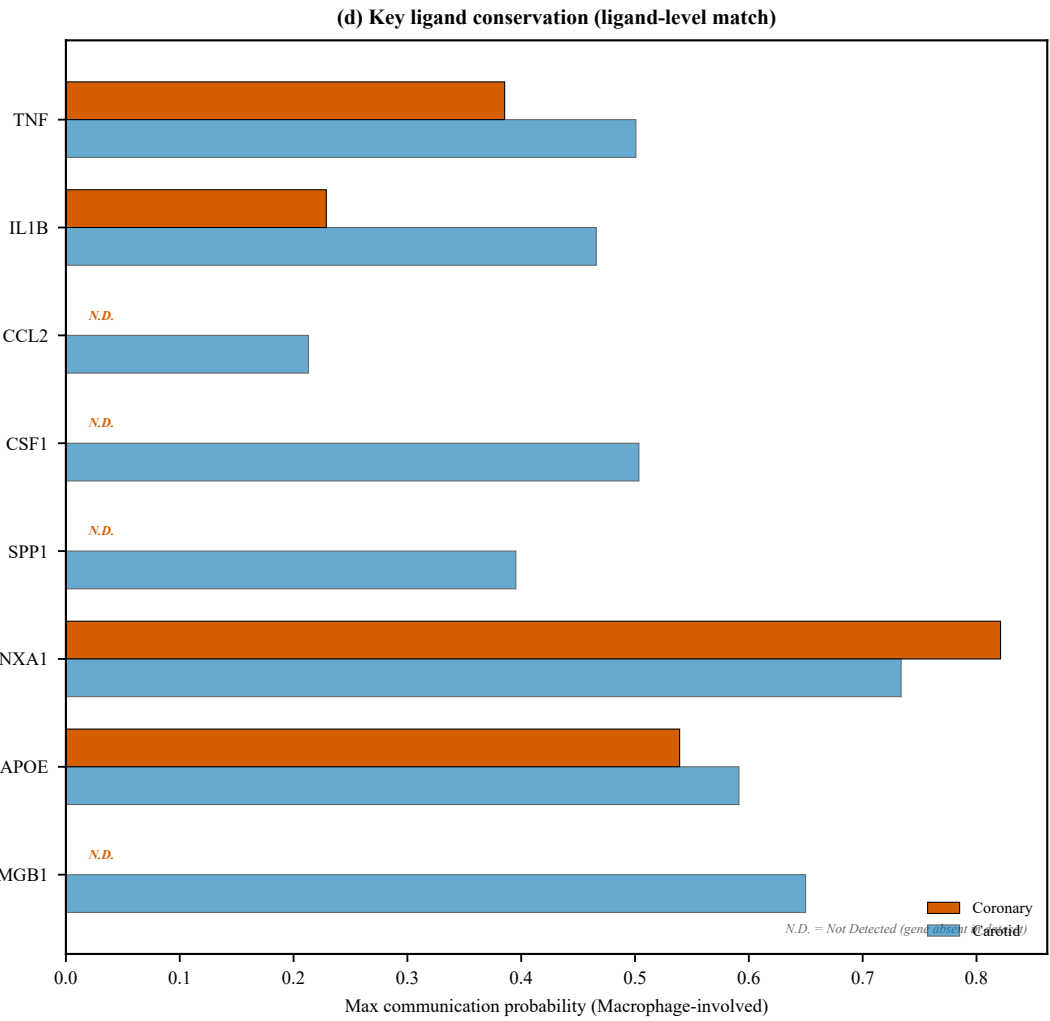
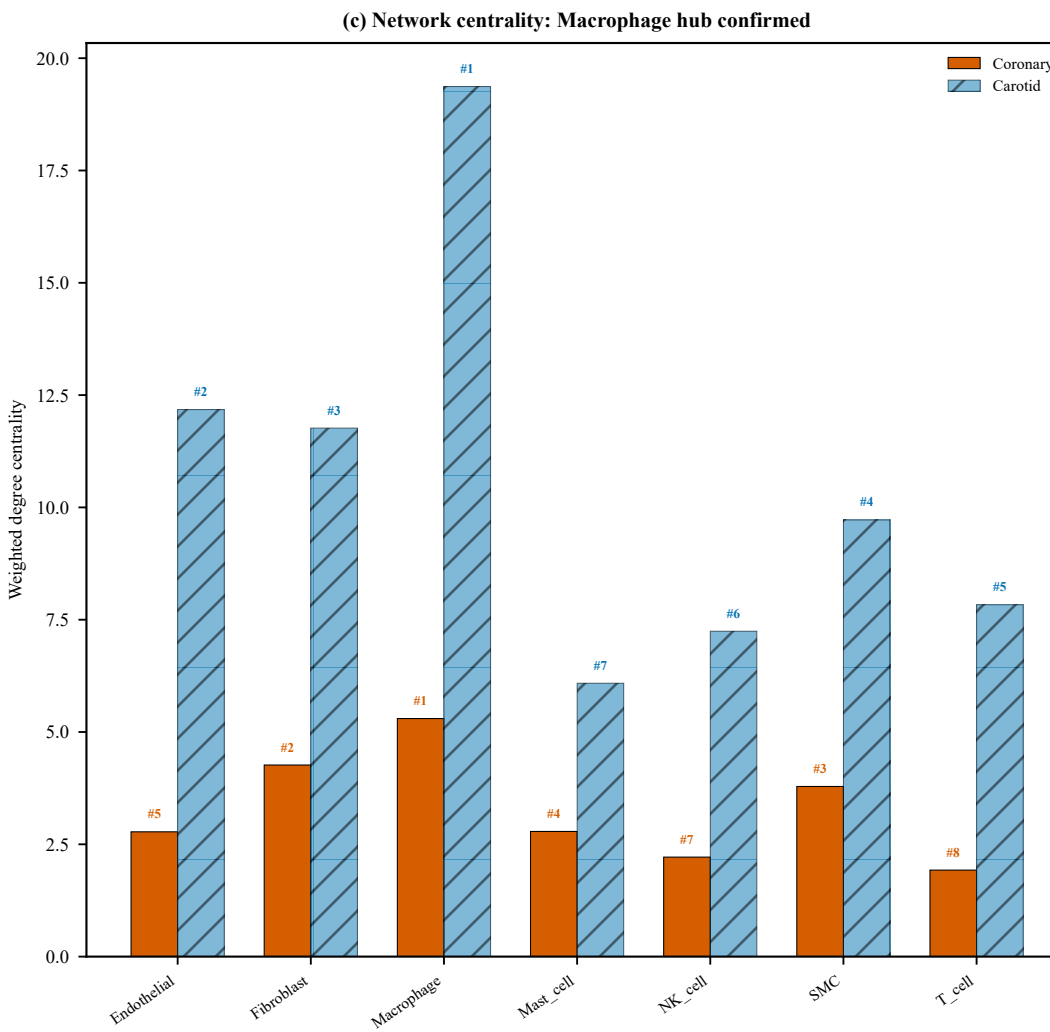
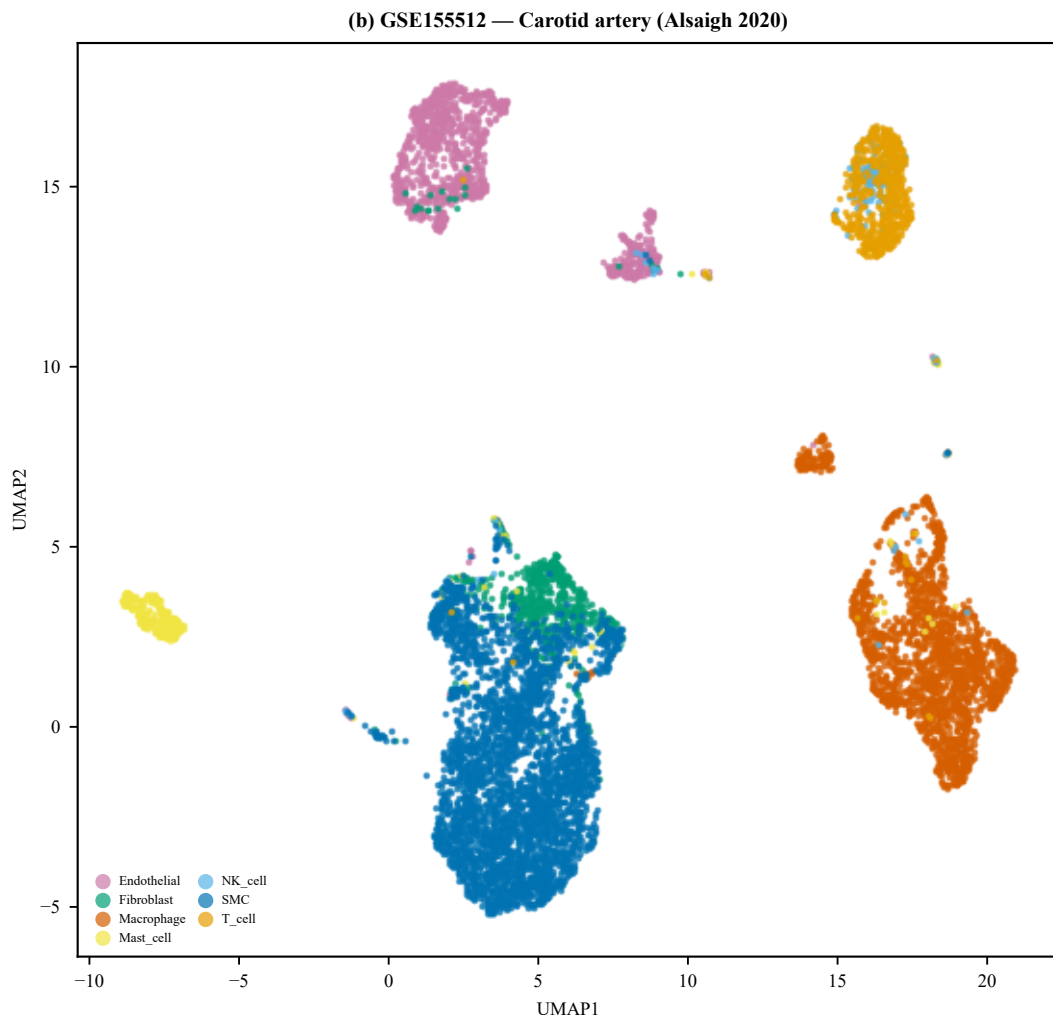
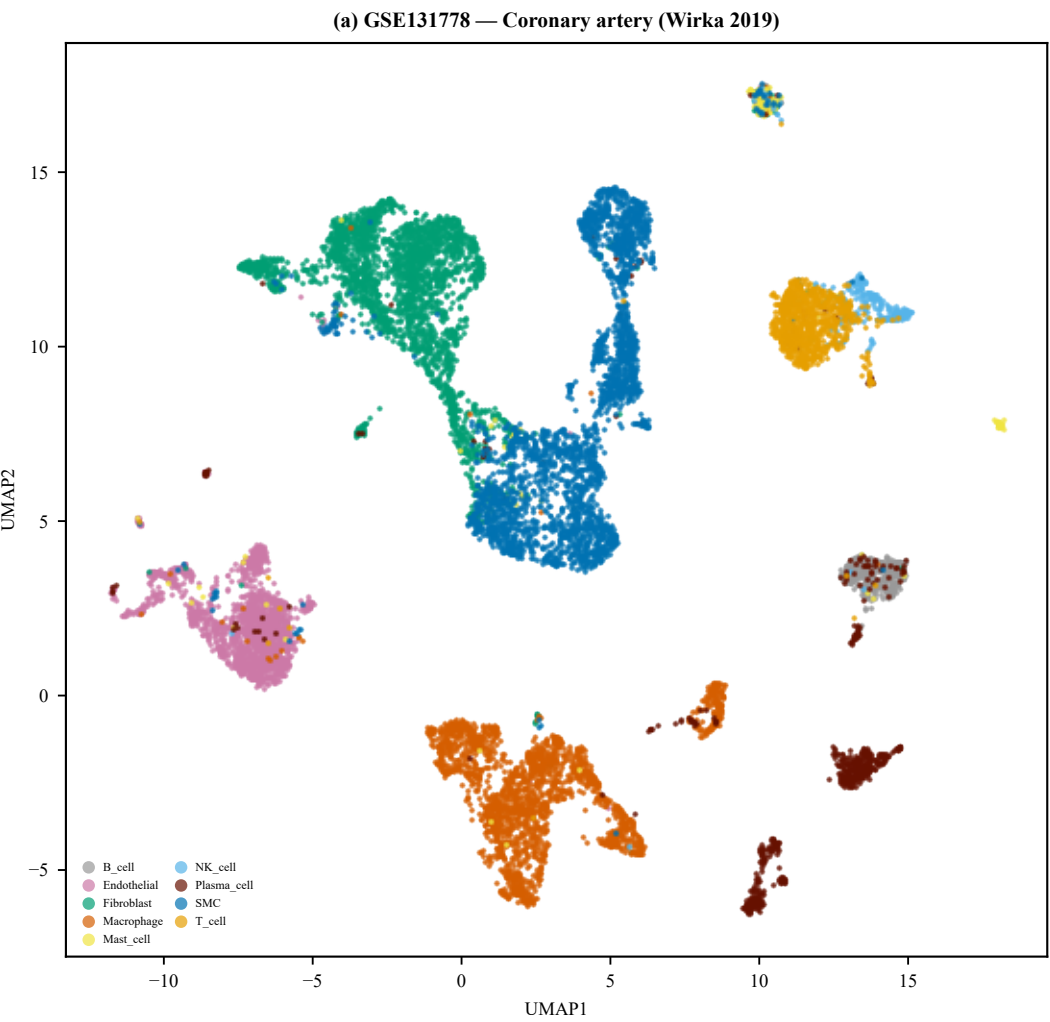


Supplementary Fig. S4 Cross-Dataset Validation of Hub Cell Mechanism (2 Independent Cohorts)



External Validation Summary

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GSE131778 (Wirka 2019): Human coronary artery
11,756 cells | 9 cell types | 2,061 macrophages
Macrophage centrality rank: #1/9 (Hub CONFIRMED)
TREMI: 23.2% Mac vs 0.4% others (near-exclusive)

GSE155512 (Alsaigh 2020): Human carotid artery
8,866 cells | 7 cell types | 2,114 macrophages
Macrophage centrality rank: #1/7 (Hub CONFIRMED)
TREMI: 16.8% Mac (conserved expression)

Cross-dataset Consensus:
Macrophage hub: conserved (rank #1 in both)
Centrality correlation: rho = 0.57 (p = 0.180)
Qualitative concordance despite conservative rank test
4/8 ligands not detected in coronary dataset
(platform/coverage difference, not biological absence)
APOE, ANXA1, IL1B, TNF: detected in both datasets

Conclusion: PLIN2+/TREMI+ hub cell mechanism
independently validated in 2 external cohorts
across 2 vascular beds (coronary + carotid).
FA0-SIRT1-MHC-II axis is vascular-bed-specific
(femoral), not universal.